Ajay Kumar Biswal

Full Stack Developer

Bengaluru, Karnataka | ajaykrbiswal@gmail.com | +91 89176 15941 | linkedin.com/in/ajaybiswalkumar | github.com/ajay-biswal | ajay-biswal-portfolio.vercel.app/

Summary

Backend-leaning Full Stack Developer with 2+ years of experience building scalable web applications using Node.js, TypeScript, React.js, and Next.js. Skilled in designing REST APIs, real-time systems with Socket.IO and WebSockets, and data layers on MongoDB, PostgreSQL, and Redis. Hands-on with authentication and authorization, API security, concurrency-safe workflows, Docker, and CI/CD.

Experience

MARAX TECHNOLOGIES | Full Stack Developer *July 2024 – Present*

- Built and maintained backend APIs and frontend modules for full-stack applications using Node.js, TypeScript, React.js, and Next.js.
- Implemented authentication and authorization using JWT, Argon2id password hashing, and role-based access control, scoped per tenant in a multi-tenant eCRM.
- Designed REST APIs and backend services with Express.js and MongoDB, improving API response times through schema design, indexing, and query optimization.
- Developed real-time features with Socket.IO and WebSockets, including messaging, typing indicators, online presence, and read receipts; used Redis caching and Pub/Sub to reduce database load and scale event handling.
- Implemented authentication and authorization using JWT, Argon2id password hashing, and role-based access control.
- Built OCR functionality in Python using Tesseract, EasyOCR, and OpenCV for automated text extraction and image preprocessing.
- Streamlined deployments with Docker and GitHub Actions, reducing manual deployment effort and improving release consistency.

Technical Skills

Languages: TypeScript,JavaScript (ES6+), Python, Java
Frontend & Mobile: React.js, React Native, Next.js, Expo, HTML, CSS, Tailwind CSS, Redux
Backend: Node.js, Express.js, NestJS, FAST API, JWT, Argon2id, Bcryptjs, OAuth, Socket.IO, WebSockets, API Gateway, Prisma
Python & OCR: Tesseract, EasyOCR, OpenCV
Databases: PostgreSQL, MongoDB, Redis
Cloud & DevOps: AWS EC2, AWS S3, Docker, GitHub Actions
Architecture: REST APIs, Event-Driven Architecture, WebSockets, Distributed Systems
Tools: Linux, Git, GitHub, GitLab, Postman, Swagger UI, pnpm, monorepo

Projects

Inventory Reservation and Checkout Service GitHub
NestJS, TypeScript, PostgreSQL, Prisma, Docker, Jest, Supertest

- Built a location-aware inventory reservation system that selects suitable warehouse locations based on serviceability, priority, and available inventory during checkout.
- Implemented transaction-safe inventory reservations and concurrent checkout protection using PostgreSQL and Prisma to prevent inventory overselling.
- Designed checkout workflows supporting idempotency, payment success/failure, abandoned and expired checkouts, with scheduled reservation expiry handling.
- Added E2E and concurrency tests using Jest and Supertest, validating reservation consistency when multiple checkout requests attempt to reserve the same inventory simultaneously.

Banter – Real-Time Chat Application GitHub
Node.js, Express, TypeScript, Socket.IO, Redis, MongoDB, Next.js, Shadcn, Lucide React

- Built a full-stack real-time chat application supporting private chats, typing indicators, online presence, and read receipts using Socket.IO and WebSockets.
- Developed and integrated frontend and backend features using Next.js, Node.js, Express, and REST APIs, with MongoDB for persistent chat data.
- Implemented PASETO-based authentication and optimized WebSocket event flow for low-latency real-time communication.
- Used Redis Pub/Sub for distributed event handling and presence tracking across multiple connections while storing chat data in MongoDB.

CRM & Real-Time Collaboration Platform | Client Project
Node.js, Express, TypeScript, Socket.IO, Redis, PostgreSQL, RabbitMQ, Docker, Next.js, Shadcn, Lucide React

- Built backend services for a multi-tenant CRM platform using Node.js, TypeScript, Express.js, and PostgreSQL, enforcing tenant-level data isolation and role/permission-based access control across APIs.
- Developed real-time team chat and customer chat with Socket.IO, using Redis [Pub/Sub / presence tracking] to support [multi-instance scaling / online status] and chat-state updates.
- Integrated RabbitMQ to handle [notifications / activity reminders / background jobs] asynchronously, [reducing request latency / decoupling services].
- Developed customer-management, task, and scheduled-activity modules end to end, from validated forms and REST API integration to tracking workflows.
- Built reusable, responsive UI components with Next.js, shadcn/ui, and Lucide React, using Redux for application and authentication state.
- Containerized services with Docker for consistent local and deployment environments.

Education

Indira Gandhi Institute of Technology | Master of Computer Applications (MCA)

- CGPA: 8.2/10
- Coursework: Data Structures and Algorithms, Computer Networks, Operating Systems, DBMS